A Survey on Sequence Alignment Algorithms and State-of-the-Art Aligners

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Vast amounts of genomic, proteomic, transcriptomics and other forms, commonly referred to as -omics data, are generated daily in an unprecedented way thanks to high-throughput Next Generation Sequencing technologies. One of the main processes to generate value and insights from this data in bioinformatics is "sequence alignment", an algorithmic routine that matches a "reference" sequence to a larger sequence. Despite the plethora of approaches, pairwise and multiple sequence alignment remain a complex problem that requires high computational power. In this article, the most prominent sequence alignment approaches of the past three decades are reviewed and categorized, examining different aspects, such as their overall algorithmic synthesis, alignment quality and performance benchmarking tests in a uniform way. The latest trends reveal an increased specialization on biology-based directions, the need for alternative heuristic approaches and a promise in optics-enabled approaches and the quantum computing paradigm shift.

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1 Introduction

In bioinformatics and computational biology, **sequence alignment** (**SA**) process is an area of high interest [1], as it plays a critical role in extracting knowledge and value from genomic, proteomic, transcriptomic and (-omics) data of vast size and scale. It is defined as the problem of optimally (partially or entirely) matching a "query sequence" e.g., of nucleotides or amino-acids, to a usually larger sequence, called the "reference sequence". In comparative genomic analysis, the aim of this process is to assess the degree of similarity between biological sequences, aiding the extraction of functional, structural and evolutionary information that can accelerate drug discovery and diagnostics. PSA is known as the problem of identifying similar regions comparing two sequences of characters. MSA is defined as the problem of comparing three or more sequences simultaneously to identify regions of similarity across all sequences. The two distinct problems of PSA and MSA are usually addressed by distinct implementations. Operations like insertion, deletion, or gap opening, take place in between sequences to optimize the similarity degree as shown in Figure 1(a),(b). The comparison may be conducted on different ranges of alignment, typically making out two modes: local SA and global SA, which are briefly described via some examples in Figure 1(c) and 1(d), respectively.

While available sequences stored in -omics databases continue to increase in size, the analysis to extract functional and structural information does not follow at an equal pace and scale, despite the need. The challenges pertaining to this problem to perform SA in scale stem from: (i) the high computational complexity, as it is an NP-complete problem, requiring significant time and resources, especially when large instances of multiple sequences are subjected for alignment [2]; (ii) the spectacularly massive extraction (generation) of biological data due to the constantly evolving **next-generation sequencing** (**NGS**) technology [3]; and (iii) the ever-growing need for faster decisions in pharmaceutics, biology and medicine [4]. Current advances in the design of efficient SA algorithms derive from accelerating research on crucial algorithmic principles and guidelines. Yet there is still a need for intelligent, alternative methods for the SA problem at scale.

In the early 1980s, sequence technology development signaled the beginning of the rapid creation of sequence databases (EMBL-Bank, GenBank, etc). Today, high-throughput sequencing technology has highly affected the way new aligners are designed. A growing number of biology applications are developed according to the "-seq" experiment [5], e.g. DNA-seq [6], ChIP-seq [7] [Park, 2009], RNA-seq [8–10], and so on. Some major paradigms are short-read sequencing and long-read sequencing. Short-read sequencing pertains to lower-cost and more sensitive analysis, while long-read is more suitable for de novo genome assembly applications.

In the main protocol of an MSA algorithm, PSA may or may not take place as an integrated sub-routine. Because of the inefficient cooperation between these two types of SA, as well as the complexity and lack of scaling of both problems, a plethora of time-efficient solving schemes have

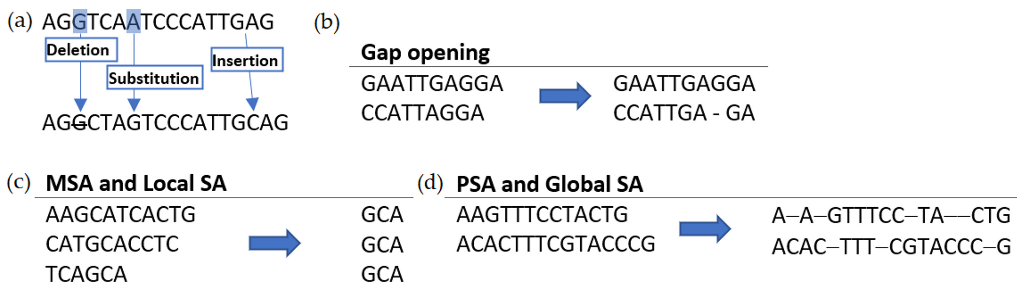

Fig. 1. Basic operations in SA procedure.

been developed during the last decade, being either heuristic or metaheuristic techniques as the most resorted approaches. Therefore, making a suitable choice for a specific application is not an easy task. Despite the various reviews of aligners currently, and to the best of our knowledge, there is a notable lack of a concentrated list of PSA or MSA tools with a description of their full protocol and their basic characteristics and algorithms. For the remainder of this article, while most detected methods are referred to as "aligners", another valid term is "tools", though it is used in a more restrictive manner. This is due to the more abstract nature of tools that could encompass web-based software or general-purpose suites, while the objective of this study is to detect methodologically sound aligners that introduce or expand SA concepts, both for PSA and MSA.

To understand and assess current and promising approaches for the future, a systematic literature review is performed [11], which is a method to collect and survey a suitable collection of scientific works for a particular research question on a specified topic. Furthermore, a bibliometric analysis is carried out on the meta-data of each selected paper, providing findings such as the most cited works, the geographical situation of the research community, the most active institutions and the temporal evolution of aligners. Aspects to be investigated include quality, replicability, reliability, and validity, including as many remarkable aligners as possible during the last three decades. The sources considered were three of the most reputed scientific databases (Scopus, Pubmed, and Web Of Science), resulting in a collection of 158 documents published in the form of journal papers, conference papers, or reviews.

In this work, a classification of PSA and MSA tools is attempted by defining a set of algorithmic stages for each SA approach and distributing the detected aligners to their respective categories according to their characteristics. Due to the wide variety of methods in some categories, more than one different staging concepts were required in order to accommodate specific groupings. Currently, no previous effort has been documented to organize and provide such an extensive pool of SA methods encapsulated under common computing stages. This fact enhances the reusability and contribution of the present work and enables new researchers to gain deeper insights into the structure of popular aligners and quickly allocate specific algorithmic characteristics. While selected prominent aligners are presented in more detail, an exhaustive list is included as supplementary material. In the closing sections, some modern approaches are discussed and the future expectations for the evolution of the field are thoroughly presented.

For the rest of this article, first, the research protocol that was followed is described, ensuring its validity. Then, the **main research question** (**MQ**) is highlighted, as well as the **secondary questions** (**SQ**) that will be answered throughout the paper. Furthermore, a bibliometric assessment is selectively presented of the collected studies (with several prominent figures illustrated in the accompanying supplementary material), highlighting trends and key points while the most prominent PSA and MSA tools are presented in a grouped way, providing in parallel their full SA protocol in a globalized approach. Finally, some future but promising methods are discussed, closing with the conclusions.

2 Research Protocol

According to the requirements of a systematic literature review [12, 13] a structured process must be employed to get a valid result. The main guidelines of the updated PRISMA 2020 [14] are adopted to facilitate a transparent and complete reporting of our systematic review. The steps in this process are specifically broken down into the following sequence: (a) establishing the primary and secondary research questions, (b) creating the search query, (c) choosing the best scientific databases, (d) setting the inclusion/exclusion criteria, and (e) reporting the survey stages.

The MQ is formulated in the following manner:

> MQ: "In the last three decades, what are the principal research works concerning the aligners addressing the SA problem?".

Several SQ are produced from the posed MQ. These SQ highlight the axis around which, scientific research has revolved around. The following SQ have been defined:

> SQ1: "Which is the temporal progression of the field under study?"
> SQ2: "Which are the main SA problems faced by the researchers?"
> SQ3: "Which type of algorithmic schemes are the most used ones?"
> SQ4: "Which are the most cited papers of the field?"
> SQ5: "Which institutes and regions are the most prolific ones?"
> SQ6: "Which are the main future challenges detected by the community?"

Writing a thorough review necessitates inclusive queries made of suitable search keywords, beyond the smart combination of the logical operators along with the topic keywords. In order to achieve this, an empirical investigation has been carried out by testing various syntax configurations. The resulting queries, that yielded the optimal number of studies, were the following two:

> Query PSA: ("SA") AND ("Pairwise") AND ("algorithm" OR "strateg*" OR "method*" OR "tool*").
> Query MSA: ("SA") AND ("Multiple") AND ("algorithm" OR "strateg*" OR "method*" OR "tool*").

The chosen queries allowed us to track and investigate all the prominent SA software aligners focusing on their algorithmic methods. The analysis of each aligner under specific criteria enabled us to provide a more detailed classification, despite the great dissimilarity among the various tools over the last three decades.

The systematic literature process is followed choosing the appropriate scientific databases. The selected databases were scientifically inclusive and contained search engines that accepted input from the established query, see Table 1.

To further reduce the extensive volume returned by the original query, further exclusion/inclusion criteria are used. The following standards were applied:

- Papers must be explicitly related to SA algorithms, methods, or processes, excluding parallel or hardware-accelerated architectures.
- The selected publications should be entirely devoted to SA, either theoretically or practically. Unrelated articles that briefly touch on these issues are not included.

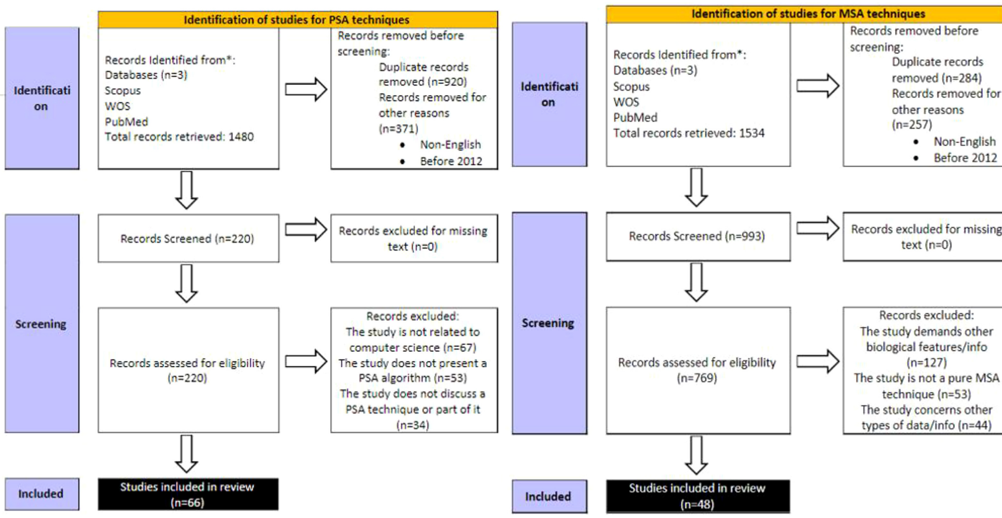

Fig. 2. Selection of studies based on the PRISMA statements, for both (a) PSA (left) and (b) MSA (right), for the years 2012–2022.

- The chosen manuscripts shall be in the English language.
- Studies on SA but of different scientific areas was excluded as were out of scope.
- We included papers that had a moderate number of citations (with a threshold of 30 citations) or that described aligners which were generally accepted by the bioinformatics community. This was carried out based on expert judgement.
- We only kept papers describing aligners that held low time or space complexity and demonstrated high performance in collective benchmarking tests.

In accordance with the outlined protocol, the following actions were carried out: (i) a search in particular scientific databases (ii) a filtering process to eliminate any papers that did not satisfy our inclusion/exclusion criteria; (iii) a manual search including eligibility criteria.

Two PRISMA statements are reported for the processing of the collected literature concerning the aforementioned two queries on PSA and MSA, in Figures 2(a) and 2(b), respectively. Given that the first organized forms of aligners started in the early 1980s, the expected literature to be extracted would be prohibitive. Thus, the literature aggregation was faced by separating the whole duration into two periods: 1980–2012 and 2012–2022. The first period was treated by using as a guide already available reviews for PSA [15–19] and MSA [19, 20–23], while the second by applying the aforementioned research protocol according to PRISMA guidelines.

3 Analysis of the Collected Publications

Following the PRISMA guidelines, the published material related to aligners, is analyzed and presented via graphical representations. While in this section we selectively present some main information about the collected studies, detailed figures are provided in the accompanying supplementary material. More specifically, SQ1 is answered in Section 3-C, while SQ2 and SQ3 are answered in sections 3-B and 3-A, respectively, along with several figures in the supplementary material. SQ4 can be found in the supplementary material while SQ5 is answered in Section 3-C and SQ6 in Section 3-D.

A Types of Aligners Surveyed

Despite the wide spectrum of applications of alignment operations in the area of Bioinformatics, the main focus of this study is on alignment software tools (aligners) for DNA/RNA, protein, SA based on their similarity. Our adopted collection of aligners is separated according to whether pairwise or multiple SA is conducted. Further clustering is imposed by classifying the aligners according to their basic protocol characteristics. The application of local or global alignment strategy is taken into consideration only for the PSA category, as MSA adopts an exhaustive approach. Additionally, our investigation is also context-aware as it concerns the aligned alphabet.

In the pairwise category, aligners of the seed-and-extend strategy are cited, making a reference to the primary **dynamic programming** (**D.P.**) and word methods. The MSA class involves aligners that belong to the known categories of star, progressive, iterative, evolutionary (genetic and swarm intelligence optimization), **Hidden Markov Model** (**HMM**)based, or **divide-and-conquer** (**DAC**) methods. Moreover, aligners that support machine learning techniques are also included.

B Types of Problems Studied

The main issues confronted by researchers in their effort to get familiarized or introduce a new alignment method are considered. In contrast to the common approach of benchmarking a specific set of aligners purely in terms of time and accuracy, a general description of their structure is presented over the most prominent aligners. In the foreground, a formulated way to present aligners is provided enabling the reader to find the type of aligner that he/she wishes to explore, and its related algorithmic strategy. The formulation is accomplished by arranging the algorithmic steps within a proposed staging which is applied for each SA method. Thus, the following fundamental challenges are summarized in the followings:

— The complexity of categorizing the wide range of alternative SA tools, including the newest ones. The decomposition and analysis of the algorithmic protocol for the so many variant aligners and make comparisons between them.
— The identification of common strategies or algorithmic combinations of already available aligners.
— The challenge of swiftly examining the attributes or features of a specific aligner. All these difficulties are addressed in the next three sections. Our systematic presentation of the software aligners is provided in compact tables for each method, at least for the most prominent aligners. A more extended list is provided in the supplementary material.

C Timeline of Aligners and Bibliometrics

The results from the adopted protocol were retrieved in November 2022, and a list was kept as supplementary material. In Figure 3, the distribution of published papers for SA software tools can be seen, divided by PSA and MSA. There is a clear dominance of MSA solutions, with 60 MSA tools being published in the last seven years (2015–2022). This coincides with the rise of several innovative techniques that allowed MSA tools to thrive and be implemented from different perspectives, such as **Genetic Algorithms** (**GAs**), Swarm Intelligence and HMMs. PSA solutions, while on the rise before 2008, have seen a decline in recent years, being outclassed by MSA tools.

The body of the study was comprised of 42 PSA tools and 116 MSA tools, with the majority of studies having been published in Journals and containing original work, or expansions of previously published work, as shown in Table 1. In addition, in Table 2 the top journals that published SA papers throughout the study period are presented. The Bioinformatics journal by

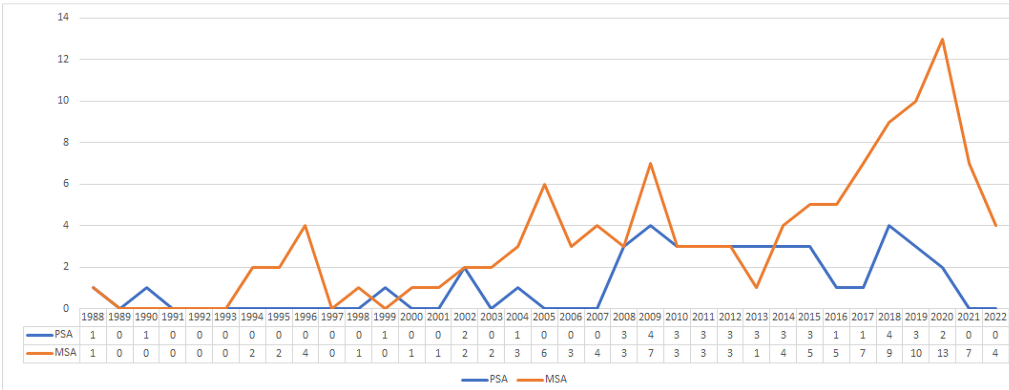

Fig. 3. Distribution of published papers from 1988 to 2022 for aligners. PSA and MSA categories are separated.

Table 1. Types of Gathered SA Studies

Type of SA	Total	Journal	Conference	Other
PSA	42	41	1	0
MSA	116	106	9	1

Table 2. Top Journals with Published SA Papers

Journal Name	Number of Papers
Bioinformatics	34
Nucleic Acids Research	17
BMC Bioinformatics	11
Nature Methods	5
Genome Research	4

Oxford Academic is generally preferred by researchers, as it is closer to the concept of SA with the rest of the journals being thematically relevant to the field as well.

A distribution of the number of papers and institutions per country in Table 3 reveals that USA is the most active country in SA research, followed by China and India. Hence, it is apparent that aligners are a research challenge of global proportions that concerns researchers from multiple global institutes. The further geographical visualization in Figure 4 shows that multiple countries from all continents have proposed solutions for SA, with Europe offering a particularly active pool of researchers and institutions, though China and the USA still hold the majority of published papers. We have to note that the assignment of countries to papers is automatically applied by distributing the paper to the country of the first author, as he/she is deemed the highest contributor to the published work. However, we acknowledge that this is a slight limitation of the country profiling.

D Open Challenges Posed by the Community

SA is the reference point of any biological macromolecular research field where PSA, and especially MSA, serve on the realization of evolutionary, functional, and structural perspective of biological macromolecules in an organized manner [24]. Moreover, high-throughput sequencing has changed the way life sciences research is conducted, due to the aforementioned numerous-seq biological applications. Finding the similarity degree in a large set of sequences is an exceptionally high-computational task while the complexity may change according to the targets of the application. In fact, the problem of computing both the optimal global alignment and the optimal local alignment for two sequences has an exact solution, either for nucleotide, or protein, sequences, but MSA applications are not efficient enough to treat them effectively.

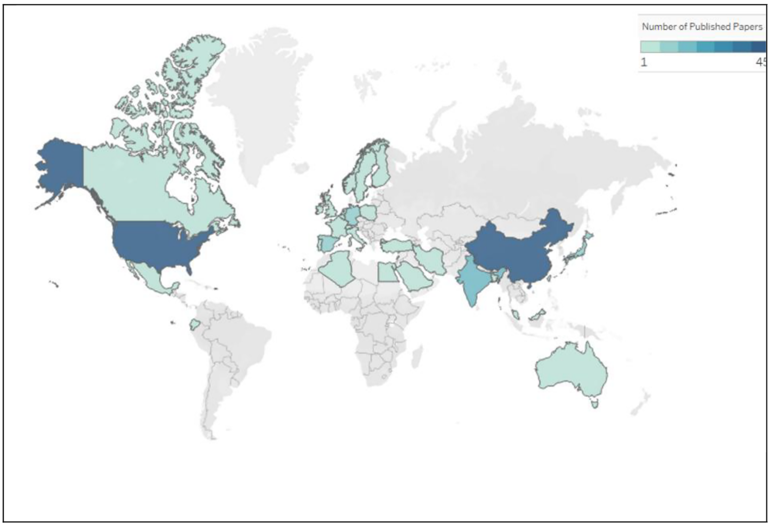

Fig. 4. Geographical distribution of researcher's country of affiliation by confirmed Papers/Institutions.

Table 3. SA Countries/Papers/Institutions

Country	Papers	Institutions
USA	101	112
China	44	44
India	20	20
Spain	11	12
Germany	13	17
Japan	10	11
U.K.	8	12
Canada	6	8
Australia	6	9
Netherlands	5	5

Another critical question for practitioners revolves around the choice of the aligner. This is not a trivial task, since there is a great variety of aligners which specialize in particular targets. Many SA tools attempt to adapt to the new technological developments [25]; the massive production of sequences; or the tackling of protocol developments like paired-end library protocols [26]. This review tries to provide a fair orientation by classifying properly the various aligners and making an analysis of the protocol of each aligner as well as presenting some characteristic and reliable benchmarking reviews, that compare aligners based on several metrics. Moreover, information about the type and size of the input data is provided as a supplementary material

In the end, is it possible to achieve further reductions in the computational complexity of the SA problem in general. From a mathematical or computational point of view, it may be carried out according to some modern concepts which are discussed in Section 6.

4 Prominent Pairwise Sequence Aligners

Pairwise alignment (**PSA**) is the foundation for many post-operative analysis on alignment procedures. A strong and efficient PSA tool will impact the final alignment result when multiple sequences should be aligned. There are two modes of such an alignment: local and global, and sometimes a semi-global alignment is observed. The first involves locating and aligning a local region, that is, similar, while the second entails end-to-end alignment. Semi-global alignment is described as an intermediate condition between local and global. More specifically, this kind of alignment provides flexibility when the terminals (either the beginning or the end, or both) of one or both sequences do not need to be matched. Such a technique provides the flexibility in handling sequences of different lengths and focusing on significant regions of similarity without penalizing terminal gaps, e.g. when one sequence is a subsequence of another or when the sequences share significant regions of similarity with differing lengths at the ends. The following paragraphs describe the timeline of PSA evolution. These selected methods are characterized by their importance and their popularity (reflected from their citations).

4.1 Dot-Matrix Methods

The first way to estimate the similarity between two sequences is traced back to 1970, when Gibbs and McIntyre [27] introduced the similarity matrix. It creates a graphical representation of the alignment providing an intuitive way to the user. This approach is conceptually simple in construction and the method itself is context-independent. Given the restricted alphabet of nucleotides or amino-acids, two sequences of the same alphabet are placed vertically on the two axis of a matrix plot and each matrix cell is marked with a dot, only if the coordinates of this cell have the same token. The construction of the matrix for long sequences is time consuming as it is proportional to the product of their length. Due to its limited capacity to encompass and analyze large sequences, dot-plotters are used mostly for visual representations.

4.2 Dynamic Programming Methods

The greatest examples of D.P. inventions are the **Needleman-Wunsch (N.W.)** [28] and **Smith-Waterman (S.W.)** [29] algorithms in 1970 and 1981, respectively. Both methods integrate a score structure that defines the function which guides the algorithms to find the optimal alignment. The reference and query sequence lengths (say L_{Ref} and L_{Query}) determine the performance of both algorithms with a complexity in runtime and memory requirements of O(L_{Ref}., L_{Query}). The S.W. algorithm aims to accomplish local alignment while the N.W. addresses the global approach. In a later time, Gotoh [30] and Altschul [31] optimized the S.W. algorithm from O($L^2_{Ref.}$, (L^2_{Query})) to O(L_{Ref}, L_{Query}) time complexity, while Myers and Miller [32] optimized it from O(L_{Ref}, L_{Query}) to O(L_{Query}) space complexity.

4.3 Word-based Methods

The first word method was introduced by Wilbur and Lipman under the title name of the FASTA program suite [33]. In contrast to D.P. methods which guarantee an optimal alignment solution, it is a heuristic method of sub-optimal alignment results but with significantly faster performance. Known also as a k-mer or k-tuple method, the concept is that some k-length words (or k-mers) of the query sequence are searched over a reference sequence. As an example, in the reference sequence "AGCT", a k-mer substring with k=2 has the following k-mers: "AG", "GC", and "CT". The choice of k depends on the application and the length of the query sequences being analyzed. Some additional processing with a cutoff scoring and narrow D.P. will lead to the final matching results.

The high-throughput sequencing technological advances created the need for further adjustments of aligners on the -seq technologies, extending word-based protocol and leading to what is known today as seed-and-extend or q-gram filter techniques. To identify the best match of the query in the reference sequence, multiple substrings of the query known as seeds, or q-grams, are generated based on the query, "anchoring" these seeds by string searching in the reference string, see the descriptions in Figure 5. The FASTA program was succeeded by the BLAST program which adopted the extended seed-and-extend word method using an estimator mechanism to evaluate the degree of similarity for each matching. BLAST soon became the most widely accepted search program for over twenty years, since the seminal publications of Altschul et al. 1990 [34] and 1997 [35]. Thus, the "expensive" brute force D.P. method was limited by the seed-and-extend protocol—excluding large areas of the reference genome.

In Table 4, we can find the most prominent aligners of this kind in chronological order. Today, they are not commonly referred to as word-based tools, but they are characterized by their seed-and-extend strategy. This category encompasses as many aligners as possible but of different characteristics, while concurrently trying to maintain the pluralistic character of this collection

Table 4. Staging and Algorithmic Protocol Analysis of Known Seed-and-extend Strategy Aligners

Year	Aligner	Method	Stages / Algorithms				
			Reference preparation	**Read preparation**	**Mapping or Seeding**	**Screening Matching Regions**	**Extension and Scoring**
1988	**FASTA [33]**	**k-mer exact**	Dot plot	Dot plot	Diagonal identification (or hashing) and scoring	Re-scoring selected regions (substitution score matrix) and rank with cutoff	(Banded) S.W. or N.W.
1990	**BLAST [34, 35]**	**k-mer exact**	Hash table	k-mer word list formation	Hashing (k-mer hash function)	High-scoring Segment Pair: These words must satisfy a score as a threshold	D.P.
1999	**MUMmer [49]**	**DAC based**	Suffix tree	–	**Long Increase Subsequence (LIS)** algorithm for sorting MUMs	–	Closing the gaps between MUMs (NUCmer)
2002	**BLAT [37]**	**k-mer exact/nearly-exact**	Hash table	k-mer word list formation	Hashing (k-mer hash function)	High-scoring Segment Pair: These words must satisfy a score as a threshold	D.P.
2009	**BWA [39]**	**MEMs**	Prefix trie (FM-index based on bidirectional BWT)	Prefix DAWG based	Backtracking (heap-like)	Overlapping criterion with penalties assignment.	S.W.
2011	**SHRiMP [46]**	**Spaced q-grams**	Hash table	q-gram index table based	Multiple grams are required for an acceptable match.	–	Vectorized SW
2011	**RUM [52]**	**Spliced-aware**	Similar to Bowtie (FM-index)	Bowtie rules	Bowtie rules and merge + BLAT rules for unmappers	Merge mappers and non-mappers.	Substitutional matrices score
2013	**STAR [48]**	**MEMs**	MMP Tree (Uncompressed suffix array)	Cluster and Stitch: Seeds clustered by proximity to anchor seeds	Maximal Exact Matches. Sequential Maximum Mappable Seed	Seed clustering by proximity and stitching.	User-defined scoring covering chimeric S.A.
2013	**BWA-MEM [44]**	**SMEMs**	Prefix trie (FM-index based on bidirectional BWT)	Prefix DAWG based	Seed and Re-seed (two rounds of SMEMs)	Chaining and chain filtering by ranking cutoff. Local and end-to-end SAs are also compared to reject	S.W./N.W. (banded affine gap penalty)
2014	**BBMAP [55]**	**Splice-aware k-mer with no count limit**	Hash table using k-mers	Read split	Hashing. Triplets alignment	Simple Score (Levenshtein distance model)	D.P.
2014	**SEGEMEHL [54]**	**Variable read length for exome data**	Enhanced suffix array (BWT).	Optional read split	–	Substitutional matrices	Bit vector-based semi-global. Re-map by lack algorithm unmappers
2014	**NOVOALIGN [60]**	**Posterior alignment probability**	Hash table	–	Hashing. Alignment quality scores (posterior SA probability)	Alignment scoring	Substitutional matrices
2018	**MINIMAP2 [59]**	**Local MEMs and Suzuki-Kasahara formulation**	Chaining process involving backtracking and k-mers algorithm	Chaining process involving backtracking and k-mers algorithm	Chain anchors identifying LMEMs. Suzuki-Kasahara formulation	Aligning short paired-end reads by a scoring function	N.W.
2019	**HISAT2 [57]**	**Hierarchical GFM index and spliced alignment**	(a) global GFM index (b) large set of small GFM indices covering the whole genome	–	Backtracking search.	Adjustable penalties	Local FM indexing for rapid extension. Non-D.P. heuristic. Substitutional Matrices Score

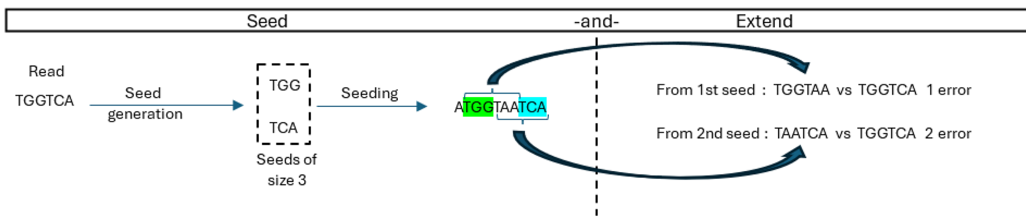

Fig. 5. The seed-and-extend algorithm. A read ATCA is sought for in GATTACA, using seeds of size two, with one error. Each seed maps once (left part). After extension of each seed (right part), it turns out that only one mapping contains only one error.

under a common staging of their algorithmic phases. The first remarkable aligners, like BLAST [34, 35], GNUMap [36], and BLAT [37], put forward the k-mer exact match seed technique. Then, SOAP [38], Bowtie [26], and BWA [39] aligners appeared supporting the k-mer inexact mode where there is a tolerance for a few mismatches without caring about their positions, in contrast to the k-mer spaced mode of RMAP [40] and Maq [41] where a specific pattern is considered with fixed wildcards. To be opposed to fixed-length seeds, LAST [42] aligner introduced the adaptive seed technique in 2011 which causes matches that are chosen based on their rareness. This technique promises a more sensitive comparison of large sequences with arbitrarily nonuniform composition. Later on, BWA-SW [43] presented another algorithm similar to **Burrows-Wheeler Aligner** (**BWA**) enhanced with S.W., to align long sequences up to 1 Mb against large reference sequences. Afterwards, the BWA-MEM [44] aligner employed the **maximal extend match** (**MEM**) seed technique in 2013, which essentially finds at each query position the longest exact match covering the position. A more accurate and efficient aligner than Bowtie2 or BWA, appeared in 2012 under the title name GEM [45]. Regardless of alignment parameters, its strategy to prune the search space without missing matches promises an exhaustive mapping with substitutions and INDELs. Actually, its filtering-based pruning scales well to the range of longer reads and this is its main attribute.

SHRiMP [46] and Hobbes [47] are known for their increased sensitivity as they make use of q-grams. Only locations that have been highly mapped by a predetermined majority of q-grams are chosen by D.P. to be aligned. The alignment procedure will ultimately decide whether or not a match is approved, acting as a filter. In q-grams strategy, the extension stage is therefore skipped. This technique has four typical stages: (a) the creation of q-grams, (b) their multiple mapping to the reference sequence, (c) the choice of the most strongly mapped regions, and (d) filtering during alignment.

Another prominent tool that appeared in 2013 was the STAR [48] aligner which manipulates spliced transcripts (RNA-seq) making use of the sequential **maximum mappable prefix** (**MMP**) seed approach, followed by seed clustering and stitching procedure. MMP is similar to the Maximal Exact (Unique) Match concept used by the large-scale genome alignment tools MUMmer [49]. The Maximal Unique Match implies that long sequences that match exactly and occur only once in each genome are almost certainly part of the global alignment. MUMmer has greatly evolved over the years from 1999, gathering a high number of citations and improving its functionality of its suffix trees and parallelism.

The reading method of a fragment generated during the sequencing procedurecan be divided in two modes: single-end or paired-end, When using single-end reading, the sequencer reads a fragment just from one end to the other, producing the base pair sequence. In paired-end reading, in the first round, reading begins from one end of the fragment till the read length is finished and then continues from the other end and it is repeated till the end if the whole fragment. An example is the GSNAP [51] aligner which can employ both modes on short or arbitrarily long length of a read. Using probabilistic models or a database of known splice sites, it may identify short and long-distance splicing in individual reads, including interchromosomal splicing.

The **RNA-Seq Unified Mapper** (**RUM**) is a pipeline based on BLAT which exploits the advantages of both genome and transcriptome mapping as well as combining the speed of Bowtie with the sensitivity and flexibility of Blat [52]. It also has a strand specific mode. Apart from alternative splicing, insertions, deletions, substitutions, sequencing errors and intron signal, RUM was tested at the base and junction levels in order to detect novel transcript features such as novel exons and alternative splicing in RNA-Seq data from mouse retina. Aliners and analyzers of RNA-Seq data like RUM has improved the annotation of the human and mouse transcriptomes. Finding novel transcripts and novel transcript variations is one of RNA-Seq analysis' most significant results, as it is crucial for the detection identification of genes harboring mutations that cause disorders and inherited illnesses.

TopHat2 is a spliced aligner [53] for long paired-end reads of RNA-seq experiments, using the Bowtie2 aligner which analyzes the mapping results to identify splice junctions between exons. Except for de novo spliced alignment, it aligns reads across fusion breaks, which can occur after genomic translocations. It also combines the ability to identify novel splice sites with direct mapping to known transcripts, producing sensitive and accurate alignments, even for highly repetitive genomes or in the presence of pseudogenes. Making use of available gene annotations, which allow it to avoid erroneously mapping reads to pseudogenes, allows to cover microexons, non-canonical splice sites, and other features of eukaryotic transcriptomes. However, TopHat2 is no longer supported and has been replaced by HISAT2.

Segemehl [54] is a prominent performer for exome data. This aligner can detect not only mismatches but also **insertions and deletions** (**INDELs**). It can accurately identify reads tainted by primer-or polyadenylation, and it is not constrained to a certain read length.

BBmap was initially designed for RNA-seq alignments in 2014 [55] but can also be used effectively for DNA as well. It is a widely used software, due to its speed and versatility, like STAR. It is a splice-aware algorithm based on the utilization of short k-mers to map the reads to the reference genome. It is an accurate aligner especially for short reads making use only of short k-mers, without size or scaffold count limit. It also exhibits higher sensitivity and specificity than Burrows–Wheeler aligners, maintainging a comparable or increased speed. It may underperform when longer reads of higher error rate are used and doesn't effectively tackle the problem of splice detection.

The first version of HISAT in 2015 [56] managed to reduce the requirements in memory running spliced alignments for RNA sequencing experiments. The first mode, HGFM, is an extension of BWT for graphs, that constructs a global **graph FM** (**GFM**) index which represents a population of genomes, together with a sizable number of local indexes (small GFM indexes) that collectively cover the entire genome. A significant notation is that aligners like HISAT2 [57] which are hashing-based aligners achieve a running time, that is, comparable to that of the quickest N.W.-based aligners, despite the fact that Hamming distance-based approaches generally have a longer runtime.

Last but not least, Minimap 1 and 2 are general-purpose alignment programs to map DNA or long mRNA sequences [58, 59]. They may perform well with noisy reads and assembly contigs or closely related full chromosomes of hundreds of megabases in length. Concave gap cost is used by Minimap2 for long INDELs, split-read alignment is performed, and new algorithms are used to lessen erroneous alignments. It outperforms the majority of aligners that are focused on a single type of alignment, being 3–4 times faster than common short-read mappers at comparable accuracy and 30 times faster at higher accuracy for both genomic and mRNA reads.

NovoAlign was introduced by NOVOCRAFT in 2014 and its evolution has provided considerable tools [60]. It is actually an effective generic pipeline supporting single-end and paired-end reads. It

Table 5. A Collective Analysis of three Comparative Studies. The "best" performers on specific indicators are shown.

Reference	Selected Aligners	Adopted Indicators	Range	Avg.	Best	Selected Datasets (Samples)
J. Kim et al. [61], 2020	[Totally: 6] RMAP, BFAST, SOAPv2, BOWTIE, MAQ, and BWA	Reads Aligned (%)	96.85–99.53	98.21	BWA	– Reference sequence NZ_CP05032 and
		Alignment runtime (m:s)	02:13–37:27	15:10	SOAPv2	– **Sequence Read Archive (SRA)** SRR11461738 of Illumina Hiseq 4000
L. Donato et al. [18], 2021	[Totally: 17] BWA, BOWTIE2, SEGEMEHL, TOPHAT2, HISAT2, MAGICBLAST, MINIMAP2, YARA, RUM, STAR, SUBREAD, BWA-MEM, GEM, NOVOALIGN, CLC GENOMICS WORKBENCH, DNASTAR LASERGENE SUITE, and BBMAP.	MAPQ ≥ [MAPQ cut-off] and AS ≥ [AS cut-off] scores	–	–	BBMAP	Simulated PE read sets were mapped to human (homo sapiens) and murine (mus musculus) reference genomes
		Mapping efficiency (tags/kb)	>90%	>99%	CLC, BWA-MEM, GEM and Magic-BLAST	– DNA: Retinitis pigmentosa WES PE – DNA: Retinitis pigmentosa WGS SE
		Mapping efficiency (tags/kb)	>45%	>99%	DNASTAR, BBMAP, RUM	– DNA: Retinitis pigmentosa WES PE
		Mapping accuracy (tags/kb)	–	~160 tags/kb	Novoalign and DNASTAR * more for DNA samples and variable for RNA	– RNA: RPE cell transcr. SE – DNA: Retinitis pigmentosa WGS SE – RNA: Osteomyelitic Exudate Transcr. PE
		Alignment runtime (m:s)	32:26–4920:00	–	Subread	– DNA: Retinitis pigmentosa WES PE
R. Musich et al. [62], 2021	[Totally: 7] BOWTIE2_E2E, BOWTIE2_Local, BWA, HISAT2, MUMmer4, STAR, and TOPHAT2.	Alignment rate distributions	23–87%	~70%	BOWTIE2_Local, BWA	48 RNA-Seq Erysiphenecator
		Runtime/read (microsec/read)	8,5–46,5	~26,30	BOWTIE2_Local	

can also report multiple alignments per read with alignment quality scores using posterior alignment probability.

4.4 Prominent Comparative Studies

The seed-and-extend strategy has evolved as the standard method in –seq technologies, and since then, most of the benchmarking studies have been focused on aligners of this category. Studying the most interesting benchmarking tests of the last five years—like those in [61–63], the conclusion is that it is not possible to define in a single way the best performing aligner, while including simultaneously all the criteria put forward. However, speed and completeness are the most acceptable indicators that all benchmarking efforts consider. The terms speed and completeness are defined as the mapping time of a single read on a reference sequence and the alignment accuracy, respectively. The second one may be viewed in terms of sensitivity, precision, or false positive percentage. Some predictions on these parameters are briefed in the following Table 5 for the best performing aligners from three comparative studies. The benchmarking workflow usually follows the next steps: (a) determination of the competitor aligners, (b) analysis of the indicators under comparison, and (c) runtime of mapping or aligning evaluations.

J. Kim et al. [61] collected six aligners for experimentation and adjusted them on three configuration modes. Despite the fact that different configurations returned different results, there were not significant performance deviations and the reported values in Table 5 are based on the default parameterization. A greater analysis was conducted by L. Donato et al. [18], where they

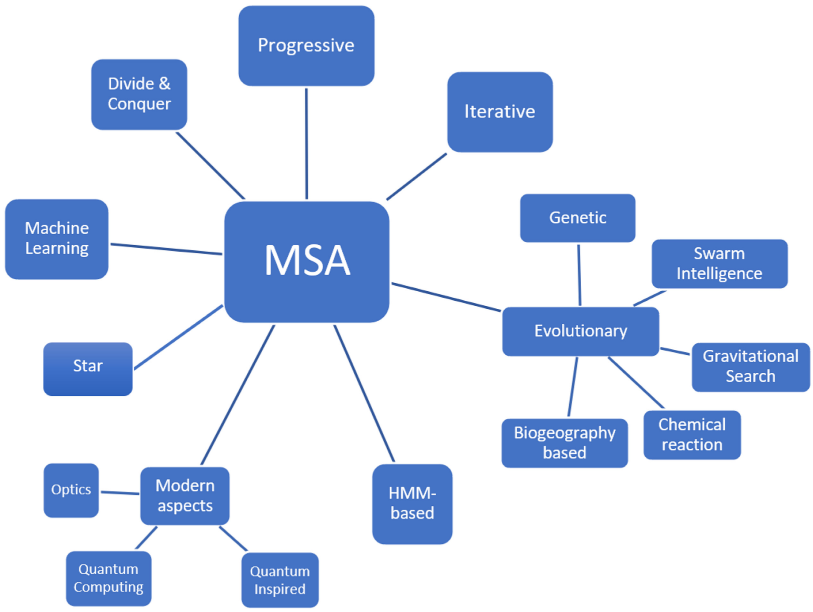

Fig. 6. Classification of MSA tools.

experimented on different aligners and tested them on simulated and real datasets using a variety of criteria. Table 5 includes the most globally applied criteria. BWA-MEM, Segemehl, Novoalign, CLC Genomics Workbench and Magic Blast are some of the best performing aligners covering most of the aforementioned criteria. In another comparative study in 2021, R. Musich et al. [62], tested seven aligners on 48 RNA-seq samples providing simulation output results on alignment rate. However, R. Musich argues that there are significant differences in this arena depending on the primary concerns of aligner-comparison.

Generally, a common assumption is that seeding is the bottleneck for short sequences while banded D.P. is the bottleneck for long sequences. As an example, BWA-MEM aligner performs better for longer reads than other popular aligners such as NovoAlign, GEM, BWA-SW, or Bowtie.

5 Prominent Multiple Sequence Aligners

In this section, the most prevalent MSA methods are presented along with their most prominent software aligners that have been developed up to the final date of collection. All MSA tools are classified under their related category and their main protocol is described under a common staging. In some categories, more than one staging approaches were employed. The classification includes the following seven types as depicted in Figure 6: Starting from the Star strategy, the progressive alignment, the iterative refinement, some evolutionary categories such as the genetic and the swarm intelligence optimization, the HMM based algorithms, the **DAC** strategy and the final category that includes MSA tools that are empowered with machine learning which imitates intelligent predictive behaviors to improve the alignment result. In each category, the staging follows the steps of the most prevalent aligner for each specific approach.

5.1 Star Alignment

Star alignment (or centre-star alignment) method is one of the oldest methods employed for tackling the problem of multiple SA, [63]. The method follows a greedy approach based on PSA and—in

Table 6. Star Strategy: Staging and Algorithmic Protocol Analysis of Prominent Aligners

Year	Aligner	Central Sequence Selection	Pairwise Alignments	Summing of inserted spaces
2015 2017	**HAlign [64]** **HAlign-II [67]**	Left-child, right-sibling creation of trie trees	Neighbour-Joining for phylogenetic tree creation	S.W. algorithm
2022	**HAlign 3 [66]**	Sequence partition and creation of k-ary trees	D.P. combined with DAG	S.W. algorithm
2017	**MASC [65]**	Ukkonen algorithm for the creation of a suffix tree	D.P. through Spark tool	Sum of scores
2022	**SaAlign [68]**	Sequence partition and creation of k-ary trees	Optimized N.W. algorithm (Spark)	Sum of scores

general—it can be described as a three-stage process, as it is described in Table 6. In the first stage, a central sequence is selected so that the "distance" between any other sequence is minimized. This can be done in various ways, such as utilization of trie trees [64] like in HAlign or the employment of a parallel framework like in MASC [65]. In the second stage, each of the remaining sequences is pairwise—compared to the central sequence, creating a "star" with the length of each spoke defining the "distance" of each sequence with the central sequence. In the last stage, the final multiple SA is extracted by subtotalling (summing) inserted spaces.

There are many variations of this method, and their common aim is the enhancement of speed by reducing the time complexity of each stage. Most of these variations utilize different approaches in order to speed up the selection of the central sequence. The most known aligners are the ones of the HAlign "family" [HA1, HA2, and HA3], with its latest addition being HAlign3 [66] following a novel approach based on the combination of K-ary tree (instead LCRS-tree), with the creation of a **directed acyclic graph** (**DAG**). One of the significant advantages of star-based aligners/algorithms is the accuracy, which comes at the cost of increased execution times [67], especially for a large set of sequences. However, in recent years, efforts were made to tackle the large dataset issue that led to the development of tools specifically designed for large datasets [68].

5.2 Progressive Alignment

The progressive class of algorithms is one of the most celebrated methods used in MSA [69]. Notably, many aligners/algorithms that originate from a different philosophy employ a progressive-inspired scheme in the final stages, in order to extract the desired alignment. The general principle of progressive alignment is heuristic in its nature and quite simple. Generally, an all-to-all PSA is created, forming a pairwise library. The library usually comes in the form of a distance matrix, that represents the relation (alignment-wise) between all possible pairs of sequences.

Then, based on that library, a guided tree is created, and a form of weighting is assigned to facilitate the next steps of the procedure. In the final stage, the most related sequences are aligned progressively, until all the sequences are aligned. In many aligners/algorithms the final alignment is obtained by combining the well-known N.W. algorithm, while others combine diverse approaches, trading execution time for accuracy and lower false-positive percentage.

There exist different aligners employing progressive approach as a foundation for their design, or as a secondary procedure. Whether we adopt the library formation, or not, the most prominent aligners of this type are presented chronologically in Tables 7 and 8, respectively. The most well-known aligners in this category are widely used and referenced such as the Clustal "tool-family" (Clustal, Clustal W, Clustal X) [69, 70], the well-known T-COFFEE [71] aligner and the TM-Align "tool-family" that specializes in protein alignment and originated from a PSA aligner, the TM-Align [72]. The rest of the family consists of MSA tools, including mTM-Align [73], r-TM-Align

Table 7. Library-based Progressive Strategy: Staging and Algorithmic Analysis of Prominent Aligners

Year	Aligner	Staging			
		Generation of primary alignments	**Processing primary alignments**	**Refining processed alignment data**	**Final Alignment**
2000	**T-COFFEE** [71]	Generation of primary libraries (Lalign and Clustal W)	Weighting aligned residues	Library extension (gap penalties applied)	SW algorithm
2005	**TM-Align** [72]	First alignment (D.P.)	Second alignment (Gapless matching)	Third alignment (D.P.)	Heuristic iteration with the help of a rotation matrix
2008	**Fr-TM-Align** [74]	Generation and scoring of aligned fragment pairs	Generation of sub-optimal alignments	Backtracking to obtain optimal path alignment	Heuristic iteration (TM-Align)
2003	**PCMA** [80]	Alignment of pre-aligned groups utilizing Clustal W	Forming of groups with low similarity	Library creation with local and global alignments	Progressively with the help of consistency **objective function** (**OF**)
2017	**HISEA** [83]	Storing reads and hashing reference set (k-mers)	Searching query set for matching k-mers	Clustering and filtering of the matching k-mers	Computing alignments

Table 8. Progressive Strategy: Staging and Algorithmic Protocol Analysis of Prominent Aligners

Year	Aligner	Staging			
		Pairwise alignment	**Distance Matrix creation**	**Guided Tree construction**	**Final alignment**
1994	**Clustal W** [69] **Clustal X** [70]	Fast approximate method or D.P.	Per cent identity scoring	Neighbour-Joining method	Alignment of larger and larger groups of sequences
2018	**mTM-Align** [73]	Modification of TM-Align PSA	Utilization of TM-Align scoring	UPGMA algorithm	NW algorithm
2020	**pmTM-Align** [74]	Data collection (Apache Spark)	PSA (Apache Spark)	Collecting results (Hadoop)	Final Alignment (OpenMP)
2005	**KAlign** [76]	Wu-Manber algorithm (pre-processing phase)	Wu-Manber algorithm (searching phase)	UPGMA algorithm or Neighbour-Joining method	NW algorithm combined with substitution matrices (BLOSUM, PAM)
2020	**KAlign3** [77]	Wu-Manber algorithm (pre-processing phase)	Wu-Manber algorithm (searching phase)	Wu-Manber algorithm (Levenstein distance for string matching)	NW algorithm
2017	**TM-Aligner** [78]	Wu-Manber algorithm (pre-processing phase)	Wu-Manber algorithm (searching phase)	Wu-Manber algorithm (Levenstein distance for string matching)	Seed alignment guides the progressive alignment
2013	**GramAlign** [79]	Implementation of relative complexity measure	Utilization of grammar-based distance estimation	Merged amino-acid alphabet	NW algorithm
2022	**LaRA2** [80]	Base-pair probability matrix	Creation of an alignment graph	Formulation of constraints into an Integer Linear Program	Utilization of MAFFT or T-COFFEE
2008	**DIALIGN-TX** [82]	Utilization of DIALIGN and DIALIGN-T procedures	Adding fragments in the alignment	Phylogenetic tree through UPGMA algorithm	Final alignment through MERGE routine
2020	**Carretta** [84]	Wu-Manber algorithm (pre-processing phase)	Wu-Manber algorithm (searching phase)	UPGMA algorithm or Neighbour-Joining method	NW algorithm
2020	**RNAmountAlign** [85]	**Posterior probability matrix(PPM)**	Refinement of PPM	Phylogenetic tree through UPGMA algorithm	Combination of progressive and iterative alignments

[74] and pmTM-Align [75] which utilize parallel computing techniques. Other tools with their own pros and cons are available, including KAlign, KAlign3, TM-Aligner, GramAlign, LaRA2, PCMA, DIALIGN-TX, and HISEA [76–83], which utilize progressive alignment as their main philosophy, but deviate significantly on the implementation of this concept. In particular, KAlign and KAlign3 [76, 77] specialize in large datasets, while TM-Aligner specializes in protein sequences. GramAlign involves an alphabet-based methodology and LaRA2 leans towards probabilistic methods. PCMA utilizes graphs and DIALIGN-TX is the latest addition of the DIALIGN "family". Another set of aligners is being developed in recent years, where the focus is not on the MSA itself; instead, these aligners focus on being available in different programming languages, like Carretta (Python) [84] or RNAmountAlign (C++) [85].

Much has been written in the literature about Clustal W and T-COFFEE and one can safely deduce that these aligners have set the par for newer aligners/algorithms. Their performance has been tested repeatedly over the years and it is quite surprising that they keep up with the competition.

The distance matrix creation is the part where many newer aligners interfere. Different approaches are used, from the employment of HMMs, [88] to optimization techniques (GAs) [86] and even Machine learning methods such as neural networks. These aligners are not included in this part, due to the significant differences between the "classical" progressive-oriented aligners that are referenced here—but the final part of the actual progressive strategy is encountered in most MSA tools and algorithms available today.

5.3 Iterative Refinement

This class of algorithms [87, 88] performs a post-processing repetitive refinement procedure, improving the overall alignment result that usually comes from progressive methods. Typically, as a continuum of a first-hand alignment, the iterative procedure provides a global alignment optimization on a given dataset by executing repetitive recalculations of their distance matrices and then the re-construction of their guide tree takes place. The guide tree dictates the merging order of sequences according to all the calculated PSA distances for all the possible sequence pairs under alignment. The iterative reconstruction of the guide tree helps the prevention of possible errors that are typically induced during progressive alignment while measuring the distance matrices or hierarchical clustering. The main ways for tree construction are the UPGMA [89] or neighbor-joining [90] methods.

The iterative behavior is determined by a conditional repetitive process which is usually defined, either by a deterministic, or by a stochastic [91] approach. For the first approach, some empirical criteria usually determine the repetition times, while for the second, some quality measures set the convergence threshold. Such criteria may be realized as an OF which is continuously optimized until it meets a high-quality alignment score. The deterministic approach may be proved sub-optimal as it may run redundant iterations, while the stochastic is self-motivated, returning different results in each iteration due to the internal use of random values. However, errors may be caused by two common traps: the known local optima problem and the moderation of gap insertion problem (or as biologists claim, "once a gap—always a gap").

Due to the wide applicability of this method, there are various aligners of this type, and therefore various strategies of selecting the sequence subgroups for re-alignment and OFs for quality convergence. The earliest MSA tool to mention this type of alignment is the MultAlign program which combines progressive and iterative refinement, developed by Corpet in 1988 [92]. After an initial distance PSA calculation for all possible pairs, a hierarchical clustering is conducted in order to cluster all similarly aligned sequences and then the construction of the tree takes place. Many similar examples shared the same concept such as Gotoh [93, 94] which employed the **random**

iterative optimization (**RIO**) using as a metric the **sum-of-pairs** (**SP**) and their **weighted SP** (**WSP**) version. Gotoh's efforts led to the publication of a doubly nested version of RIO method by using WSP under the title name PRRP [95]. This aligner uses a hill-climbing algorithm to optimize its MSA alignment score [96] and iteratively corrects both alignment weights and locally divergent or "gappy" regions of the growing MSA [94]. The alignment of individual motifs is then achieved with a matrix representation similar to a dot-matrix plot in a PSA.

Other popular MSA tools of similar logic that make use of tree rebuilding are MAFFT [97], MUSCLE [98], and PRIME [101]. Most of them, refine a progressive alignment by dividing it into two groups horizontally and then realign the two groups with another round of progressive alignment, until the overall alignment is converged. MUSCLE was invented by Edgar in 2004, achieving a profile-based fast and rigorous group-to-group alignment at each iteration minimizing the overall computational cost. MAFFT aligner managed to further speedup the process by utilizing Fourier transform in 2002. Various benchmarking tests have shown a satisfactory trade-off outcome between time consumption and alignment accuracy. In 2005, PRALINE [102] provided a homology-extended strategy with profile pre-processing and the secondary structure-guided scoring scheme strategies which can be set to use consistency information to drive subsequent alignment iterations. This secondary structure prediction significantly improved the alignment quality. A summary of all these techniques is encapsulated under a common staging in Table 9.

Although most of the iterative aligners are compatible with the proposed staging, a second one (see Table 10) was needed to the steps of SATé and its subsequent evolution. In 2009, the meta-method SATé [103] made use of the affine gap penalty to a phylogenetic analysis program (POY) in order to improve alignment quality, while guide tree consistency is supported by the highest-scoring maximum likelihood method. The next version SATé II [104], in 2012, further improved their centroid decomposition technique offering better performance and accuracy. PASTA [105] is based on SATé but adopts a parallel strategy to support large scale alignments.

5.4 Evolutionary Methods

In recent years, the rise of evolutionary algorithm methodologies to perform SA has been observed. [21, 23]. Evolutionary algorithms employ stochastic techniques, such as GAs and swarm intelligence approaches in order to complete the process of SA.

GAs [106] are widely used in finding the best solutions in chromosome encoding and follow a philosophy that favors the survival of the optimal chromosomes and DNA sequences. GAs fall under the category of metaheuristic approaches and follow the principles of natural selection in order to produce an optimal outcome.

The second pathway of evolutionary algorithms aimed at performing SA via protocols related to swarm intelligence. At their core, swarm intelligence algorithms, with **Particle Swarm Optimization** (**PSO**) [107] being a characteristic representative, aim at spreading particles in a given space by maximizing a given measure, which acts as an OF.

5.4.1 Genetic Approach. The concept behind GAs is that the process of MSA runs iteratively, producing a new generation of alignments in each iteration. The first stage of the GA process revolves around preparatory stages in the alignment, ensuring that the inputs have proper representation and format. In the second stage of the GA, an initial population, usually comprised of alignments that are passed as inputs and are represented in a specific way, is created by using specific algorithmic techniques.

This initial population is then evaluated based on an OF, which is the third stage, and serves as a criterion of fitness for the optimality of the produced generation. The best fitted alignments of the initial population are then selected and subjected to crossovers, mutations and other operators so

Table 9. Iterative Refinement Strategy I: Staging and Algorithmic Analysis of Prominent Aligners

Year	Aligner	Cycle	Initial SA	Clustering	Guide Tree Construction	Progressive SA	Scoring	Iteration Condition
1988	**MultAlign** [92]		Distance matrix calculation (FASTA).	By PSA scoring.	According to clustering.	Group-to-group scoring.	Update hierarchical clustering	Repeat if new clustering differs
2002	**MAFFT** [97]							
	Mode 1: (Optional: Progressive+)	1st	Distance matrix based on the number of k-mer shared by a pair of sequences.	–	Using distances calculate with a modified UPGMA with modiefied linkage.	Approximate Group to group alignment by AFFT (FFT-NS-1 or NW-NS-1).	–	–
		2nd	–	–	Reconstruction.	Re-alignment (FFT-NS-2 or NW-NS-2).	–	WSP score convergence or cycles # reaches 1,000
	Mode 2:		Distance matrix based on all PSAs.	–	Using distances calculate with a modified UPGMA.	Approximate Group to group alignment by AFFT (G-INS-1/L-INS-1).	–	Repeat till WSP + Consistency scores COFFEE convergence.
2004	**MUSCLE** [98]	1st	k-mer Distance Matrix.	–	Guide Tree 1 (UPGMA).	MSA1 list.	%ids calculation from MSA1.	–
		2nd	Kimura Distance Matrix calculation from MSA1.	–	Guide Tree 2 re-construction (UPGMA).	MSA2 realignment list.	Split tree2 into two subtrees	–
		3rd	–	–	–	Re-align profiles.	SP scoring.	Repeat from tree splitting.
2006	**Probalign** [99]	.	Partition function and posterior probabilities.	Probabilistic consistency transformation.	Align sequence profiles along a guide-tree (UPGMA).	Pairwise HMM.	–	–
2020	**ProgSIO-MSA** [100]	.	Progressive SA.	Look Back Ahead with improved SP.	–	–	Position-Residue-Specific Penalty.	SIO.
1996	**PRRN/PRRP** [95]		Given list.	By PSA scoring.	According to clustering.	WSP optimization.	Nested iterative refinement of MSA set.	Repeat till convergence (doubly nested).
2005	**PRALINE** [102]	.	Profile homology for each sequence (PSI-BLAST).	Secondary structure-guided or transmembrane-guided strategies.	–	Exchange weight matrices.	By using DiAlign2 or other global strategies. Apply gap penalties.	Repeat till e-value cutoff.

that the next generation is produced, and the population is updated. The process is continued until either a predetermined number of generations is reached or if other termination conditions are met.

It should be noted though that, while the aforementioned stages are the traditional GA approach for the MSA problem, it serves as a baseline categorization. Many of the introduced aligners that use GA in their methodologies involve several substages and different algorithms, based on their architecture, concept and design. However, the described stages indeed portray the GA process in a satisfying way and all aligners can be roughly categorized under them. In the

Table 10. Iterative Refinement Strategy II: Staging and Algorithmic Analysis of Prominent Aligners

Year	Aligner	Cycle	Initial SA	Clustering	Guide Tree Construction	Centroid Decomposition	Progressive SA	Iteration Condition
2009	**SATé [103]**	1st	Initial MSA list with RAxML on each alignment	–	GTR+Gamma ML tree construction with best score (POY).	CT-k decomposition on the tree is employed.	Re-alignment of produced subsets using MAFFT	–
		2nd	–	Subsets are merged using MUSCLE.	GTR+Gamma ML tree construction using RAxML(nonoverlapping subsets).	–	–	–
2012	**SATé-II [104]**	1st	Initial MSA list with RAxML on each alignment	–	ML tree construction.	MSA list is divided into two sets by removing the longest branch in the tree (recursed til each subset has 200 taxa).	Re-alignment using only the RAxML tree in MAFFT	–
		2nd	–	Subsets are merged using Opal.	ML tree construction.	–	–	–
2015	**PASTA [105]**	1st	Backbone alignment with HMMER (an HMM is employed).	–	FastTree-2 construction using the simple profile HMM.	As Sate II, tree split into disjoint subsets.	–	–
		2nd	–	–	Spanning tree formation for each subset.	–	Subset sub-alignment using MAFFT L-INS-I.	–
		3rd	Final alignment with transitivity merge on trees	PSA merge of aligned subsets.	Tree estimation using FastTree-2	–	–	–

following paragraphs and table, some notable aligners are presented. A full list can be found on the supplementary material.

There are multiple aligners for MSA that utilize GA and variations, as presented in Table 11, with many of them being developed in the last 10 years, showcasing the tremendous rise in popularity of these types of algorithms in MSA [108]. One of the most prominent tools is SAGA [109]. SAGA is one of the first aligners that uses GAs to perform MSA. It introduces the concept of initial population generation and breeding for subsequent generations, based on the evaluation of a fitness function. It remains one of the most widely used aligners that utilizes pure GA for the performance of MSA. GA is generally regarded as a reliable and efficient technique for MSA. Mo-SAStre [110] is another known aligner that uses Pareto Optimal Scoring and mutation/crossover operators to perform MSA. In addition, VDGA [111] applies Vertical Decomposition to evaluate the generated population and update the alignments accordingly by combining previous generations. RBT-GA [112] integrates the GA with the Rubber-Band technique and performs MSA by using Elitism and selecting the best individuals. Finely, MSAGMOGA [113] and BSAGA [114] use variations of the GA algorithm from a Multi Objective and a Bi-Objective perspective to improve upon the concept of SAGA, with the former using similarity and affine gap penalties for fitness evaluation and the latter uses the TCC score and merge gaps to accelerate the process.

While these aligners are considered as the prime examples of MSA performed by GA, recent years have seen the introduction of other techniques. GAPAM [115] is an aligner that combines Progressive alignment with GA and uses Weighted sum of Pairs to evaluate fitness while

Table 11. GAs: Staging and Algorithmic Analysis of Prominent Aligners

Year	Aligner	Staging				
		Preparatory Stages	**Initial Population Creation**	**Fitness Evaluation**	**Population Update**	**Termination**
1996	**SAGA [109]**	Alignments with terminal gaps	Initialization based on OF	Probabilities, Genetic operators (crossovers, gap insertions, block searching, and local rearrangement	Crossover - Mutation	100 generations
2013	**MO-SAStre [110]**	Alignments from other tools	Mutation and crossover operators	Pareto fronts and NSGA-II algorithm, STRIKE Score, No Gaps Percentage, and TCC	Additional mutation and crossover	N number of generations
2011	**VDGA [111]**	DP Distance Table, Preconstructed Guide Tree	DP distance table, Kimura distance table, Sim-of-pairs, and Operators	Vertical Decomposition Algorithm	Combine generations for new population	100 generations
2009	**RBT-GA [112]**	Sequences in 0/1 format	Crossovers and mutations, Rubber Band Technique, and Genetic Operators	Elitism Technique	Iterative until OF is maximized	Predetermined number of iterations
2014	**MSAGMOGA [113]**	Sequences in specific format	Field weighting	Similarity, affine gap penalty	Crossover – Mutation operators	100 generations
2010	**GAPAM [115]**	Alignments from other tools as input	Mutation operators	Single point crossover, multiple point crossover, mutations, and WSOP	50-50 split	100 generations
2019	**GA-CRO [116]**	Importing of sequences with normalized lengths	Random Non-Repeating Permutations	Tournament Selection, SinglePoint Operator	CRO Algorithm, RecombineMatchColumn Operator, CollectSpace and CatchMatchColumn Operator	Decomposition Threshold
2016	**GAPWA [117]**	Alignments from other tools as input	Random Permutations and Matrix DP Scoring with Traceback	WSOPS	Selection-Crossover-Mutation	50 generations

GA-CRO [116] relies on the principles of **Chemical Reaction Optimization** (**CRO**), paired with GA. Finally, GAPWA [117] leverages GAs in conjunction with D.P., namely the Needleman Wunsch algorithm, in order to accelerate the mutation and crossover stages.

5.4.2 Swarm Intelligence Approach. Another promising evolutionary algorithm approach is swarm intelligence. The aligners that fall under this category perform the MSA process based on swarm optimization algorithms, which simulate the behavior of particles in a given space. As mentioned above, one of the most famous swarm intelligence algorithms is PSO [107]. The principal concept of the swarm intelligence approaches is the dispersion of several individuals (referred to as particles) in a given space, with the ultimate goal being to assign all particles in the optimal positions. The particle positions, location and velocity are crucial factors to deduce their movement in the examined space, while an evaluation (or fitness) function is used to determine the optimal path for each particle. A typical swarm intelligence algorithm, hence, involves the initialization and placement of the particles in the examined space, the evaluation and update of their position, the calculation of the fitness of each particle and finally the particle movement that changes their position in the space and updates their parameters (velocity, location, position). Usually, a swarm

Table 12. Swarm Intelligence: Staging and Algorithmic Analysis of Prominent Aligners

Year	Aligner	Staging					
		Particle Creation	**Position Evaluation**	**Position Updates**	**Fitness Evaluation**	**Particle movement**	**Termination**
2018	**EGSA** [137]	Binary Alignment	Total force, acceleration, and velocity	Gravitational Constant	WSOP	Binary GSA	Optimal positions
2017	**BFO-GA** [125]	GI	Bacteria behaviour algorithms	Selection, Crossover, Mutation Operators	Similarity, Gap Penalty, and Non-Gap Percentage	Swimming and Tumbling	Optimal positions
2018	**PSOSA** [140]	SA as particles	Point crossovers and Local Shuffle operator	Gap addition	SOPS	SAn	Optimal positions
2017	**MOAFS** [129]	GI and MAA	Optimal Pareto Solution	Mutation and Crossover Operators	WSOP, Similarity	BlockMove operator	Optimal positions
2020	**WOAMSA** [130]	Random initialization	SOSS	Prey encirclement algorithm	SOPS	Shrinking and spiral updating algorithms	Optimal Positions
2016	**ABC_Aligner** [122]	Bee agents, temperature, and Cycles	Food sources evaluation based on local information	Workers and Onlookers and Scouts Updates	SOPS	Get optimal path	Termination on max cycles
2021	**DSPSO** [141]	Particles as vectors, particle generation by gap insertion	Crossover Operators	DP	SOPS	Path Selection based on probabilities	Optimal Positions
2018	**ABC-BFO** [125]	Insert Unaligned sequences, Randomize positions	Food sources evaluation	Workers and Onlookers and Scouts Updates	Similarity, Gap penalties	Swimming and Tumbling	Optimal positions
2018	**BL-DPSO** [142]	Position matrices, velocity matrices, personal best matrices, and global best matrices	GI	Update velocity and position for two levels	Fitness value of particle positions	Move global best of swarm	Optimal positions
2018	**MSA_CRO** [134]	Random Matrix Initialization	On-wall Ineffective Collision, Inner-Molecular Ineffective Collision	Decomposition, Synthesis	Forward Repairing, Left Right Adjustment	Update thresholds	Decomposition Threshold
2016	**HMOABC** [124]	Random Selection	Fast Non-Dominated Sorting	Rank and Crowding Distance	Mutation Operators	Local Search	Fast Non-Dominated Sorting
2022	**ICROMSA** [135]	N.W. algorithm	SOSS	Decomposition, Synthesis	SPS Scores	On-wall ineffective and Inter-molecular ineffective evaluation	Optimal Solutions
2016	**IBBOMSA** [139]	BB Habitat Representations	**Habitat Suitability Index (HSI)**	Immigration and Emigration rates	HSI	Migration and Mutation Operators	Optimal positions and Termination Conditions

intelligence approach in MSA and in other problems is terminated when the optimal positions have been reached for all particles.

There are several algorithms that are being utilized in MSA and follow swarm intelligence methodologies, as showcased in Table 12. One of the most famous is the **Ant Colony Optimization** (**ACO**) algorithm [118] which simulates the movement of ants and the discovery of optimal paths for colony movement based on pheromone levels. Several aligners proposed for MSA utilize the ACO algorithm, representing sequences and part of sequences as ants that seek the optimal path, which in the case of MSA is the discovery of the optimal alignment. One of the most prominent aligners that utilizes this algorithm is GA-ACO [119],. This aligner effectively uses genetic operators and ACO to measure the probability of survival for each ant agent that selects a specific path and outputs the optimal positions of particles (or ant agents), providing the best solutions for alignment of sequences. Another well-known aligner is AntAlign [120], which uses probability matching and BLOSUM matrices to evaluate path selection and provide robust solutions to MSA.

Another popular algorithm that belongs to the swarm intelligence concept is the **Artificial Bee Colony** (**ABC**) algorithm [121], which simulates the foraging processes of bees, that operate like a hivemind when seeking optimal food sources. The concept of the algorithm is the definition of three bee populations (workers, onlookers, and scouts) that each play a different role in locating potential food sources for the hive. The objective of the algorithm is to simulate the communication between these different populations when evaluating the food sources and output the best ones. ABC_Aligner [122] is considered the most prominent aligner in this category, implementing the complete process of the algorithm to produce the optimal food sources, which in the case of MSA are optimal solutions for aligning sequences. This aligner uses the Sum of Pairs Score as an OF for food source evaluation and follows the exact steps of the ABC algorithm. Important aligners as well are ABC_SA [123] which follows similar pattern with ABC_Aligner and HMOABC [124] that utilizes fast non dominated sorting to and a novel aligner called ABC_BFO [125] using **Bacterial Foraging Optimization** (**BFO**) [126], another PSO algorithm that simulates the behavior of bacteria dispersal in organisms. The behavior of bacteria foraging is also leveraged in BFO-GA [127], and in TS_BFO [128] using penalty functions and TabuSearch to enhance the discovery of motifs in sequences.

Other algorithms that are used in aligners and resemble the swarm intelligence approach simulate the behavior of fish swarms [129] and whale populations [130], using Pareto Optimal Selection and prey encirclement processes. In addition, other approaches include Partial Order alignment, integrating sequences in a graph representation [131, 132], CRO [133–135] that uses decomposition and synthesis procedures and treats alignment as a chemical reaction that transforms molecules into products and Gravitational Search [136–138] that use gravitation and Newtonian laws of motion to calculate particle velocity and provide feedback on alignments. Last but not least, **Biogeography-Based Optimization** (**BBO**) is another interesting swarm-based methodology that simulates the concentrations of particles as biological habitats, measuring habitat suitability and updating particle velocities as they move to different biogeographical regions [139].

Finally, several aligners [140–142] use the classic PSO algorithm with variations, offering a Two-Level approach that executes two concurrent PSO procedures for optimal alignment, a D.P. variation and a combination of PSO and Simulated Annealing, where temperature is seen as an OF for alignment evaluation.

It is worth noting that other swarm intelligence techniques such as Bat and Firefly Optimization [143–145], Frog Leap Algorithm [146–148] and Cuckoo Search [149] can also potentially be used for MSA and the exploration of their capabilities is an intriguing research direction.

5.5 HMM Based

Models based on HMMs or HMM-based models are a class of probabilistic models that rely on the creation of a Markov process [150]. In particular, most HMM-based algorithms constitute a three-stage procedure, with the first stage being the preparation stage. In most models, this concerns the creation of an all-to-all pairwise PPM. The second stage, which is the training stage picks up right

Table 13. HMM-based Algorithms: Staging and Algorithmic Analysis of Prominent Aligners

Year	Aligner	Initialization	Final Matrix computation	Clustering or Guided tree	Final Stage
2018	**Clustal Omega** [162]	Initial alignment, distance matrix (k-tuple), and guided tree	Refined distance matrix (HMMER)	Refined guided tree (HMMER)	Use of HHAlign to refine alignment
1996	**HHAlign** [155]	Initialization of the dataset	All-by-all PSA	D.P. (Forward algorithm)	Progressive Alignment
2005	**ProbCons** [157]	PPM creation	Probabilistic Consistency Transformation	Hierarchical Clustering	Progressive Alignment with SP scoring
1996	**HMMER** [168] **PFTOOLS** **HMMPro**	Calculation of emission and transmission probabilities	PPM	Creation of the Guided Tree	Viterbi / Forward algorithm
2010 2019	**MSAProbs** [158] **ProbPFP** [159]	PPM creation and Partition Function	Distance Matrix and sequence weighting	Weighted Probabilistic Consistency Transformation	Progressive Alignment + 10 iterations
2021	**ProPIP** [86]	Implementing **Poisson Indel Process** (**PIP**)	D.P. under PIP	Forward phase	Alignment through Backtracking
2017	**M2Align** [161]	Insertion of pre-computed MSA solutions	NGSA-II algorithm (Initializing population)	NGSA-II algorithm (first alignment)	Alignment is refined via STRIKE
2020	**Sequoya** [160]	PyMSA for scoring an MSA solution	NGSA-II algorithm (Initializing population)	NGSA-II algorithm (first alignment)	Alignment is refined via iterations

where the preparation stage ended. The PPM is manipulated in order to lead to the creation of a hidden (non-observable) sequence and on a symbol (observable) sequence. The hidden sequence of states is the reason these models are called "hidden". The final stage is the alignment stage, i.e. the extraction of the MSA, combined with some sort of evaluation about the final alignment.

For the alignment stage, most HMM-based models rely on an iterative and/or progressive procedure which depends on the findings of the training stage. Despite the evident memory requirements, the advantage of HMM-based models is the fact that they get more accurate as the number of sequences-to-be-aligned grows, which is a direct consequence of the type of their probabilistic architecture. Specifically, for all Markov processes (which includes HMM), the probability of achieving the exact alignment is maximized as the number of samples (sequences) grows.

However, it should be pointed out that the three-stage process is a rough categorization, and in most aligners, each of those stages consists of many sub-stages, which in some cases employ different techniques (e.g. GAs, Bayesian statistics) [151–154] or utilize different parts of other software (e.g. PFTOOLS) [155, 156]. This is done to enhance the accuracy of the models, usually at the cost of execution time. In recent years, the space requirements, remain fairly the same since in most models, the "newer" posterior probability matrices replace the old ones.

Currently, there is a significant variety of HMM-based aligners, organized and presented in Table 13. In earlier years, pure HMM models were created (HMMER, HHAlign) [150, 155] alongside other probabilistic models (e.g. ProbCons, RNAmountAlign) [85, 157]. These aligners, despite their differences mostly on the preparation stage, employ the same architecture and subsequently, their performance does not vary significantly.

However, the development of optimization techniques led to the creation of hybrid HMM models where some part of the procedure is refined by combining the HMM with a partition function and/or an evolutionary algorithm. Such examples are three aligners with significant differences, yet similar philosophy: MSAProbs (partition function) [158] ProbPFP (partition function and PSO) [159], Sequoya (a Python-based software aligner, employing the NGSA-II evolutionary algorithm)

Table 14. DAC: Staging and Algorithmic Analysis of Prominent Aligners

Year	Aligner	Staging			
		Sequence input or seeding	**Processing input or seeding**	**Transitory stage**	**Final alignment**
2020	**GSAlign [168]**	Utilization of Burrows-Wheeler transform (suffix tree)	Seeding by identifying **Local Maximal Exact Matches (LMEMs)**	Choice of the alignment procedure	Gapped or un-gapped alignment
2021	**MAGUS [169]**	Centroid sequence decomposing of SATé-II	Aligning subsets by utilizing	Graph Creation by ProbCons (MPC)	Markov Clustering and criteria satisfaction
2021	**Super5 [170]**	Centroid sequence decomposing of SATé-II Query Anchors	Guided tree by using UPGMA algorithm optimal chain	Cluster alignment via Multi-threated ProbCons (MPC) Break Sequences	Progressive alignment (via MPC profile)
2022	**FMAlign [171]**	Build Indexing common seeds between sequences	Find optimal chain	Parallel Alignment	Generate entire alignment via MAFFT, Clustal Omega, and so on.
2020	**FAME [172]**	Finding common seeds between sequences	Remove common noises and create common roots	Building compatible roots into a chain via a DAG using a known MSA-Aligner	Generate overall alignment via MAFFT, Clustal Omega, and so on.
2020	**SpliVert [173]**	Vertical splitting of the MSA in three parts	Removing gap characters of the middle part using a known MSA-Aligner	Realigning of the middle part using a known MSA-Aligner	Splice the middle part with the other parts

[160] and M2Align (a popular JAVA-based software aligner, similar to Sequoya) [161]. Another interesting case is the Clustal Omega [162]. It is the latest offspring of the well-known Clustal "tool family" and employs a HMM to increase accuracy: in particular, it utilizes aspects of HHAlign.

A crucial point is that in most HMM aligners and algorithms, the final (alignment) stage is achieved through a "classical" algorithm (e.g., Viterbi) combined with D.P. for alignment scoring, gap penalties, splice junctions, and so on. Usually, iterative alignment or progressive alignment or their combination is encountered in existing solutions [163, 164]. However, in recent years, there has been a development of aligners that utilize different techniques for the alignment, such as the creation of a (DAG – MAGUS+eHMMs) [165] or a combination of HMM with progressive alignment and optimization techniques (ProPIP) [86].

5.6 Divide-and-Conquer

The **DAC** philosophy is one of the oldest and most efficient ways to derive and effectively implement any type of algorithmic procedure. In general, the first stage concerns the "breaking" of sequences into smaller sub-sequences. In this stage (which usually involves many sub-stages), the sub-sequences are aligned and the final MSA is produced by merging the sub-sequences alignments into an overall alignment. As it is custom in MSA, especially in the recent decade, there is a growing number of aligners that combine different techniques. One of the advantages of this philosophy is a certain degree of independence between stages. This fact enables the combination of different techniques, to enhance accuracy, execution time, memory usage or space requirements.

The most representative aligners of this kind are summarized in Table 14. A prominent class is composed of aligners based on the Seed-Chain-Align philosophy (MiniMap2, GSAlign) [56, 168]. These aligners are quite often referenced in the literature and despite their differences, they are based on short-reads, which additionally are "chained" together with the help of "anchors", until an alignment process extracts the final MSA. For the final stage, “classical” algorithms can be employed, usually with the help of D.P.. Other aligners, divide horizontally the sequences into sub-sequences (PASTA, MAGUS, and Super5) [105, 169, 170] or vertically using common seeds (FMAlign, FAME) [171, 172]. Super5 utilizes a HMM for the middle stage and shares some common aspects with ProbCons. The horizontal division leads to the alignment of short-length sequences that can be accurately aligned faster than longer sequences, while the vertical division exploits optimal chains (FMAlign) or DAGs (FAME), in order to concatenate the results, before the final, alignment stages.

All these aligners utilize a scoring scheme (usually SP combined with gap-penalties), while the final extraction of the MSA is achieved with the help of a classical algorithm (progressive, iterative and/or combination). Notably, these aligners rely on other tools with the same philosophy to obtain intermediate results, e.g., FMAlign uses Minimap2 to obtain the optimal chains, while BWT is often employed in the beginning stage.

PASTA is based on the horizontal division of sequences, by computing an initial tree. Then, it iterates between estimated alignments and tree estimations. MAGUS is based on PASTA but deviates substantially on the alignment stage. More specifically, MAGUS utilizes the HMM to create a library which leads to the creation of a DAG. In particular, MAGUS+eHMMs employed a specific set of HMM (e-HMMs) to improve both accuracy and execution time of an already fast and accurate aligner (MAGUS). Another aligner requiring specific mention is SpliVert, [173].; its purpose is to refine MSA. In particular, after the aligning tool has obtained an initial alignment for a given set of sequences, SpliVert splits it vertically into three parts. Then, it removes from the second (middle) part, all gap characters and uses another MSA-aligner to realign that part. Finally, it splices together the two remaining parts, with the realigned middle part, to obtain a refined alignment.

5.7 Machine Learning

Recent years have seen the rise of techniques for MSA that rely on Statistical Computation and Machine Learning. The concept behind these approaches is that Machine Learning classifiers are trained on existing alignment datasets or on alignment probabilities and are then utilized to predict future alignments for optimal alignment scores. The construction of the classifiers is evaluated with different OFs, either by calculating posterior probabilities, tuning the parameters and comparing the results with typical MSA algorithms (e.g. N.W.) or by using convolution, to name a few.

The most notable examples are presented under a distinctively staged format in Table 15. RAWMSA [174] is one of the most popular aligners for Machine Learning based MSA, using Deep Learning and exploiting a two-layer Neural Network for predicting alignments. The prediction is achieved with the use of Long Short-Term Memory and Bidirectional Recurrent Neural Networks, using SoftMax and 2D Convolution for evaluation. The final output of the aligner is the sequences from the different network layers. DRL_MSA [175] is another Deep Learning aligner that uses LSTM Networks and Q-Learning for training and evaluation, producing aligned sequences as a result. SPARKMSNA [176] is a distributed solution that uses partitioned dataframes to represent sequences and a supervised learning classifier to construct a suffix tree of aligned sequences. BayesCAT [153] and GISMO [177] use Bayesian Logic and Monte Carlo Markov Chains to infer the probability of two sequences having overlapping regions, with the former using Markov Models and the latter HMMs for training. The outputs are the distribution of posterior probabilities, in the case of BayesCAT or clusters of correlated sequences in the case of GISMO. Finally, RAlign [178] uses Reinforcement Learning, with the Asynchronous Advantage Actor Critic algorithm used for training and a Convolutional and a Feedforward Neural Network used for prediction. The output of this tool, apart from aligned sequences, are the probabilities of each state of learning which are then utilized to enhance predictions in future alignments.

Table 15. Machine Learning Augmented: Staging and Algorithmic Analysis of Prominent Aligner

Year	Aligner	Staging				
		Preparatory Stages	**Training**	**Prediction**	**Evaluation**	**Output**
2019	**RAWMSA** [174]	Embedding with Trained two-layer NN	2D Convolution	LSTM, BRNN, SoftMax, and 2D Convolution	2D Convolution	Outputs from Network layers
2019	**SPARK-MSNA** [176]	HDFS and RDDs for suffix tree construction	Pattern matching with Length differences	Supervised learning layer	Store inputs from previous steps	Form final sequence with suffix tree
2018	**BayesCAT** [153]	Indel Model Construction with MCMC	Markov Models	Calculate posterior probabilities	Evaluate Posterior Distribution	Posterior Probabilities
2019	**DRL_MSA** [175]	Memory capacity initialization	Weighting	LSTM network	Q-learning, Q-network	Aligned Sequences
2016	**GISMO** [177]	Random Alignment Creation	HMM	MCMC	Bayesian Integral Log-odds	Clusters of Correlated Sequences
2019	**RlAlign** [178]	Initialize parameters (neurons, filter size, reward states, and steps)	A3C	CNN, FNN	Hyperparameter tuning and comparison with NW	Aligned Sequences and state probabilities

5.8 Prominent Comparative Studies

In Table 16, the most characteristic benchmarking reviews of MSA tools that utilize different datasets comprised of sequences are presented along with their evaluation metrics, such as the BaliScore and the TC-Score, as well as the execution times.

Given the great plethora of MSA tools, someone could consider alignment operation from many aspects following the classification provided. Moreover, the need for benchmarks was demanding in order to provide a better orientation for practitioners. A seminal benchmarking among popular aligners was conducted in 2011 [162] following the standards of BAliBASEv3 [179] that utilized the PREFAB and HomFam darassets for additional comparisons. In this framework five groups of references, containing 82, 41, 30, and 16 alignments, were included.

The MSA problem can also be translated as a multi-objective optimization problem. B. Chowdhury et al. [21] evaluated the performance of GAs in a comparative study between CLUSTALW/X and a set of GA-based aligners like MSA-GA, SAGA, RBT-GA, GAPAM, and VDGA. The benchmarking tests ran on hand-curated datasets of BAliBASE and were executed for five different reference groups of sequences with different characteristics. The benchmarking process showed that every GA-based algorithm performed better than CLUSTALW/, with different aligners performing better on each reference group.

R. Alvaro et al. [180] provided three different multiobjective characteristic-based algorithms for increasing alignment accuracy and conservation for a given set of aligners by finding their best possible parameterization. The adopted aligners were the well-known Kalign, MAFFT, MUSCLE, MSAProbs, PASTA, Clustal Ω, ProbAlignm ProbCons, and T-Coffee. They proved that their multiobjective approaches can make optimal decisions on the configurations of the aligning parameters to improve the SA quality. All the competitors were tested on three different benchmarks on large datasets evaluating their performance by calculating SP and total column scores.

T. Paruchuri et al. [23] made a fair attempt to address the question if it is possible to treat the MSA problem using evolutionary computation techniques like GA combined with PSO, **gravitational search algorithm** (**GSA**), BBO and CRO. Different stopping criteria may be applied being either fixed or reaching optimality. The implementation of four algorithms on GA and PSO, GSA, BBO,

Table 16. MSA Benchmarking Reviews with Key Points

Reference	Selected Aligners	Adopted Indicators	Range	Avg.	Best	Selected Datasets
B.Chowdhury et al. [21], 2017	[Total: 8] CLUSTAL X, CLUSTAL W, MSA-GA* SAGA*, RBT-GA*, GAPAM, VDGA, and MO-SAStre	BaliScore (SP and Total Column Score) ranging from 0–1	Group 1: 0.042–0.993	Group 1: 0.744	Group 1: SAGA	5 groups of sequences contained in BaliBase 2.0
			Group 2: 0.154–0.997	Group 2: 0.647	Group 2: RBT-GA	Group 1: 18 datasets of equidistant sequences Group 2: 23 datasets of highly divergent sequences
			Group 3: 0.130–0.918	Group 3: 0.598	Group 3: MO-SAStre	Group 3: 11 datasets of sequences with 25% similarity
			Group 4: 0–0.865	Group 4: 0.343	Group 4: MO-SAStre	Group 4: 2 datasets of sequences with N/C terminal extensions
			Group 5: 0.422–0.852	Group 5: 0.716	Group 5: GAPAM	Group 5: 2 datasets of sequences with insertions/deletions
T. Paruchuri et al. [23], 2022	[Total: 4] GAPWA GSAMSA IBBOMSA ICROMSA	BaliScore	Group 1: 0.4823–0.9834	Group 1: 0,770	Group 1: ICROMSA	3 groups of sequences contained in BaliBase 2.0 Group 1: 15 datasets of equidistant sequences
			Group 2: 0.6423–0.9945	Group 2: 0.896	Group 2: ICROMSA	Group 2: 23 datasets of highly divergent sequences Group 3: 11 datasets of sequences with 25% similarity
			Group 3: 0.315–0.9352	Group 3: 0.715	Group 3: ICROMSA	Group 4: 10 datasets of sequences with N/C terminal extensions
			BaliBase 3.0 Reference	0.851	IBBOMSA	Group 5: 5 datasets of sequences with insertions/deletions Additional comparisons with BaliBase 3.0
			BaliBase 3.0 BB	0.704	ICROMSA	with same Groups
A. Rubio-Largo et al. [180], 2018	[Total: 12] MAFFT, MSAProbs*, ProbAlign*, ProbCons*, T-Coffee*, Clustal Omega, PASTA, MUSCLE, KAlign, fwk-MAFFT, fwk-MUSCLE, and fwk-KAlign	Q-Score, Runtime, TC-Score (on SABmark and HomFam)	17.56–95.39%	62.4%	fwk-MAFFT	PREFAB v4.0 groups of different sequence identities, 1682 total datasets
			03:02–29:04:24	7:33:19	KAlign	
			22.16–39.24%	31.11%	fwk-MUSCLE	SABmark v1.65
			6.28–14.11%	10.25%	fwk-KAlign	
			00:00:03 – 00:13:08	00:01:56	fwk-KAlign	
			40.14–91.17%	77.04%	fwk-MAFFT	HomFam dataset 3 groups of small, medium and large sequences
			002:47:11 – 152:21:01	79:02:48	fwk-MAFFT	
			16.75–81.66%	59.89%	fwk-KAlign	
F. Sievers et al. [162], 2011	[Total: 12] MSAProbs*, ProbAlign*, MAFFT, ProbCons*, Clustal Omega, T-Coffee*, KAlign, MUSCLE, FSA*, Dialign*, PRANK*, and ClustalW*	BaliScore, Running Time (on PREFAB and HomFam)	0.220-0.865	0.497	MSAProbs	Balibase 3.0 dataset with 5 different families with 218 datasets
			0.277-0.980	0.664	Clustal Omega	PREFAB v4.0
			00:01:21-97:38:18	19:40:42	KAlign	
			0.216 - 0.708	0.482	Clustal Omega	HomFam
			00:03:58 - 79:38:31	13:08:34	MAFFT	

Note that aligners marked with * do not participate in all tests.

and CRO (GAPWA, GSAMSA, IBBOMSA, ICROMSA, respectively) took place for the conduction of a BaliBASE benchmarking test under the estimation of a global fitness function. The results showed that the GA based solutions were relatively computationally expensive due to their iterative behavior and it is supported that the final result depends on various factors like sequence length, number of sequences, the encoding scheme or the quality of the fitness function. However, these approaches are viable working efficiently in terms of accuracy and time for solving MSA problems.

The final alignment quality achieved by different software aligners may be differently affected, not only by their adopted alignment algorithm, but also by the kind of the adopted quality estimation method. J. Chao et al. highlights the discrimination between the structural benchmark or simulated sequence datasets and those methods that are based on common strategy characteristics. Structural benchmarking may provide a restricted applicability using fixed test sets on protein structures (e.g. BAliBASE [179], Pfam [181], and HOMSTRAD [182]), while simulated sequences adopts a probabilistic model with configurable test sets which may include some deficiencies (e.g. ROSE [83], INDELible [184], and Dawg [185]). Scoring sum of pairs or true column scores constitutes the most common ways to estimate the similarity of two alignments by counting the number of common pairs and common columns in both alignments. Ranking tests may also estimate the p-value (e.g. Friedman and Wilcoxon [186] tests) to assess acceptable error deviations. The other way, with a commonality-based approach (e.g. MUMSA [187]) employs multiple overlap [188] or head-or-tail [189] scoring methods, while datasets have no dependencies on specific models. It is based on the concept that more than one alignment algorithmic protocols are better than a single, but this logic may lead to mistakes.

6 Discussion and Promising Approaches for the Future

Large SA datasets may have a significant scale disparity between each alignment as compared to small-scale and simple data in most bioinformatics fields. The constant and rapid accumulation of extracted sequences in biological databases have exposed SA machinery to super-scale sequences, which poses a great challenge for the current SA algorithms. The design of a method able to handle super-scale sequences while assuring accuracy and efficient execution times presents a significant difficulty. As an example, since the Covid-19 pandemic emerged, the number of available related genomes has grown dramatically in public repositories, but they are prone to low quality with sequencing/assembly errors, thus making the accurate re-alignment of all genomes nearly impossible on a daily basis [190].

Higher expectations for correctness-aware aligners are imposed by the imperfect biological sequence analysis scenarios. The majority of MSA techniques have difficulty in accurately predicting the evolution of sequences due to hardware limitations imposed by current technologies and the absence of an accurate evolutionary model [191].

Some alternative directions have been raised to improve the algorithmic complexity of SA methods with the majority of them focusing on time or memory space complexity contributions. These directions concern the evolutionary algorithmic approach by exploiting some quantum inspired approaches or even pure quantum computing techniques. Advancements in optics and some interesting proposals may also provide a different approach.

6.1 Quantum Inspired Approaches

The first approaches [192–198] to combine quantum-based techniques with evolutionary or GAs were proposed between 2005 and 2010. The core idea is expanded over three steps: encoded initialization of the population, application of quantum operators to properly evolve the overall encoded state towards the desired alignment state, and measuring the final state, in order to extract the optimized result from qubits. The whole procedure may be repeated until a criterion is satisfied or a pre-specified number of iterations is completed. Quantum computation relies on concepts like superposition of states, qubit representation, and quantum parallelism where huge amounts of quantum states can be processed concurrently and in parallel by a quantum computing system.

From 2010 and onwards, the metaheuristic **quantum-behaved particle swarm optimization** (**QPSO**) [199, 200] is used to address the computationally hard training of profile HMMs, providing optimal alignment of multiple sequences. In QPSO, each single particle is treated as a spin-less one moving in quantum space. Moreover, given the analysis of QPSO, the **diversity-maintained QPSO** (**DMQPO**) [201] includes a diversity control strategy into QPSO to enhance the global search ability of the particle swarm. Experimental results have demonstrated that the benchmark datasets perform better when HMMs were trained using DMQPSO and QPSO than the other common classical techniques like Baum-Welch and PSO. Generally, the core idea consists of: (a) initialization of an array of particles with random position and velocities inside the problem space; (b) determination of the mean best position among the particles; (c) evaluation of the desired OF (for example, minimization) for each particle and comparison with the particle's previous best values; (d) Determination of the current global position (minimum) among the particle's best positions. (e) The current global position is compared to the previous global: if the current global position is less than the previous global position; the global position is placed of the current global; (f) For each dimension of the particle, a stochastic point is adopted attaining the new position by stochastic. Once again, the whole procedure is repeated with convergence on particular stop criteria.

6.2 Quantum Computing Approaches

Some new ideas have been reported in the late five years based solely on purely quantum computing algorithms. The first way is based on modified versions of the well-known Grover's algorithm. In [202], a recurrence plot technique is employed to be combined with a known quantum multi-pattern recognition method when the recognized patterns exceed the limit of the 1/3 of the overall patterns. In 2021, a more comprehensive method [203] realized on quantum indexed bidirectional associative memory uses approximate pattern-matching based on Hamming distances. After embedding properly biological sequences into qubits, this algorithm allows approximate matches and distributed searching. The second way is highly based on a modified version of R. Schützhold's algorithm appeared in 2019 which is mainly based on the idea of quantum pattern recognition [204]. A dot-matrix method is combined with a quantum pattern recognition algorithm to simplify the detection of similarities between sequences. All these methods claim a significant complexity improvement on SA.

6.3 Optics Approaches

Another class of approaches having the potential to produce concrete results in multiple SA problems, stems from the field of optics, such as digital signal processing techniques [205] or moiré matching [206]. The base of these approaches is the **Discrete Fourier Transformation** (**DFT**) and the consequences of the widely known correlation theorem. It should be noted that the **Fast Fourier Transformation** (**FFT**) used in MAFFT and other aligners, is in fact an algorithm implementing DFT on large sets of sequences. Thus, approaches modeled by means of DFT present a natural relationship with MSA techniques [207].

In digital signal processing techniques, each of the four DNA/RNA nucleic acids (A, C, G, and T) is represented by a number which corresponds to a gray-level pixel. Thus, the genome sequences can be represented as images consisting of four gray-level pixels, enabling the implementation of the highly sensitive correlation methods. Necessary for the implementation of this approach is the use of an optical correlating architecture [205, 208]. Furthermore, the use of neural networks [209] enhances the effectiveness of these methods, while more advanced optical correlators can further improve the overall performance [210]. Despite promising results, these methods suffer from the necessity of complex optical structures which restrict their applicability.

Moiré techniques are quite popular in 3-D sensing, due to its ability to measure the shape of 3-D objects fast and accurately. A typical architecture includes two gratings and a light source, but digital pattern projectors have been utilized recently and robust methods, which simplify hardware demands while improving accuracy are developed [211]. Their implementation in MSA consists of the visualization of the genome sequences, by employing different color coding (binary, hydropathy, polarity) enabling the identification and localization of patterns. The problem of moiré techniques lies in the fact that only a pair of sequences can be compared at a time, but this can be overcome by employing a multiplexing coloring technique [212].

7 Conclusions

Overall, SA plays a critical role in many aspects of modern genomics research, being a powerful tool that helps scientists to better understand the complexity of living systems. Some specific applications are the realization of the evolutionary relationships between different organisms, the identification of functional regions of a molecule such as genes or regulatory elements, and the prediction of the structure and function of a protein. Additionally, SA can identify potential targets for drug or diagnostic tests development.

The software development of PSA and MSA tools is evolving rapidly as every year modern solutions appear in this field. According to the latest reviews speed and completeness co-exist in opposition rendering difficult the integration of all the benefits in a single aligner. Furthermore, the frequent and constant adaptations to new dictations or requirements make the implementation and the consolidation of a standard evaluation method or benchmarking protocol a challenging task. Each aligner is oriented towards particular requirements that may be useful for a particular context. Moreover, one requirement may negate the others, not allowing their coexistence in a single aligner.

A wide classification of a thorough list of aligners is demonstrated linking the software tool with its algorithmic protocol including pairwise or multiple SA. Most of the previous reviews collect only specific software characteristics or deal with just a restricted group of aligners, running benchmarking measurements on accuracy and speed. The adopted datasets are usually completely different depending on the test, raising doubts about the performance results. Surely, it is a difficult task to provide more accurate benchmarking estimations due to the polymorphism of these aligners.

The milestone aligners have been presented, providing practitioners with easily accessible ways to compare different algorithmic protocols or strategies and come up with the most suitable solution. Despite the difficulty to concentrate and organize information about the various significant aspects and qualities of the aligners, the adopted staging managed to extract common denominators for each category, substantiating a systematic approach and a broader collection view.

High-throughput NGS is continuously advancing offering profound technology which creates new horizons for the development of new applications of clinical use. NGS can be used to sequence entire genomes or constrained to specific areas of interest. There is an on-growing list of countries [213] which have developed programs to sequence large populations. It is important to organize and support the manipulation of all these ventures with the proper computing framework.

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